ESP32 Gas Leakage Detection System



MQ Gas Sensor
Gas Leak Detected

ESP32
Microcontroller



Buzzer
Alarm ON



Servo Motor
Close Gas Valve



Exhaust Fan
Fan ON



LED Status Indicators

SAFE
Blinking Green
Signifies Safe

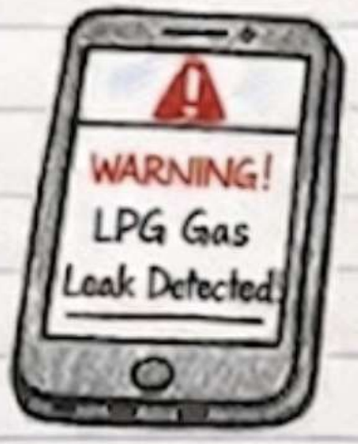


UNSAFE
Blinking Red
Signifies Unsafe





Wi-Fi Module



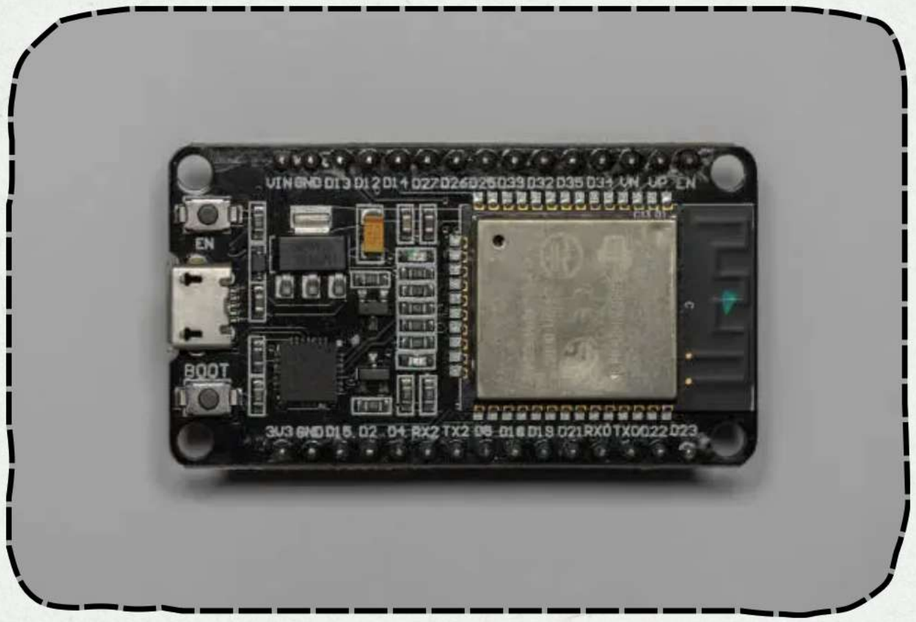
Mobile App Features:

Real time data measurement
Control of Exhaust fan and servo motor functioning
Alert to Phone

Major Components Used



Detects LPG concentration in surrounding air and converts it into an electrical signal for the microcontroller.

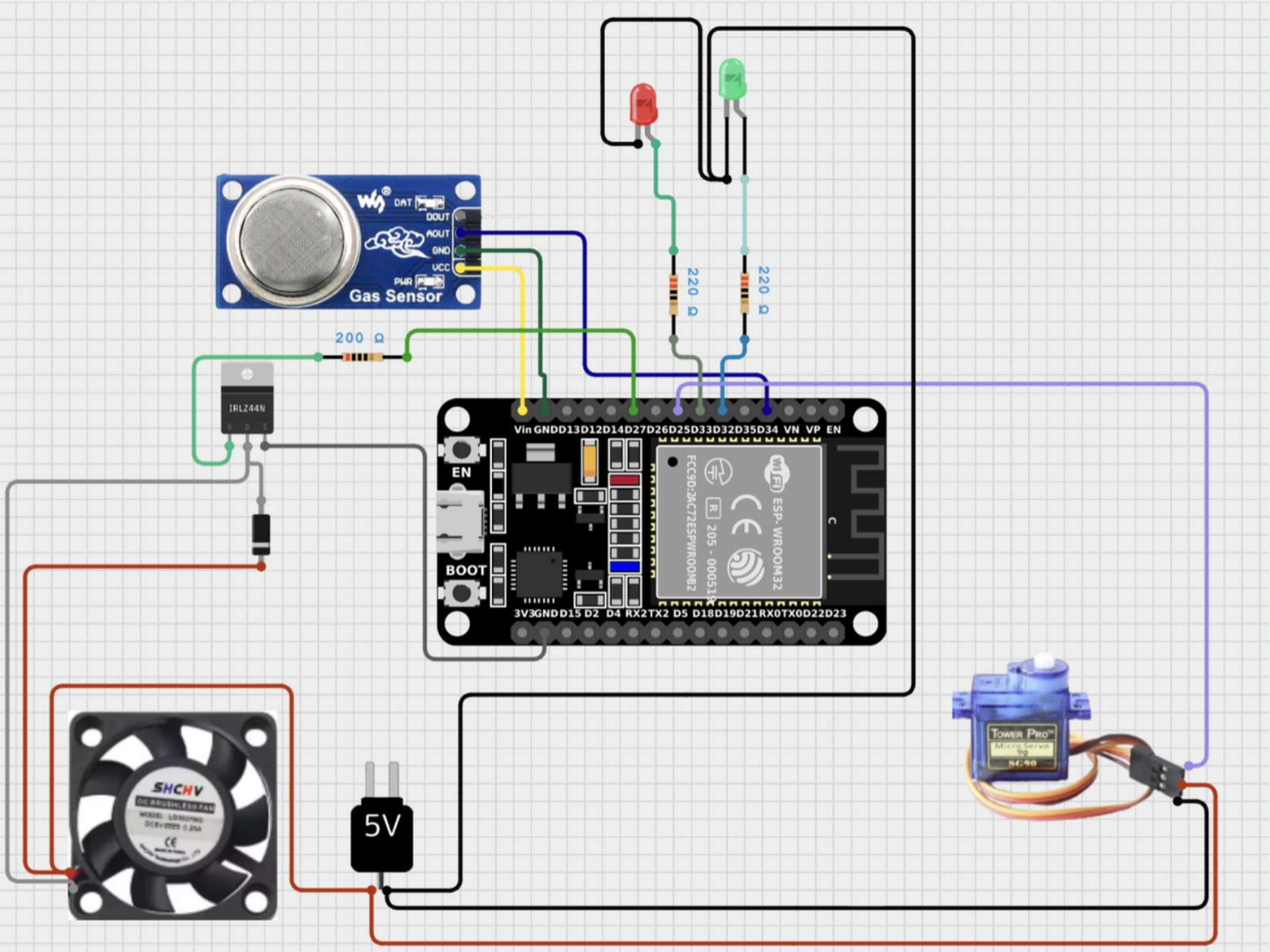


32-bit dual-core processor with high clock speed, multiple GPIO interfaces, analog-to-digital converters, and integrated Wi-Fi/Bluetooth connectivity.
Supports real-time data processing, low power consumption, and reliable communication for scalable embedded safety monitoring systems.



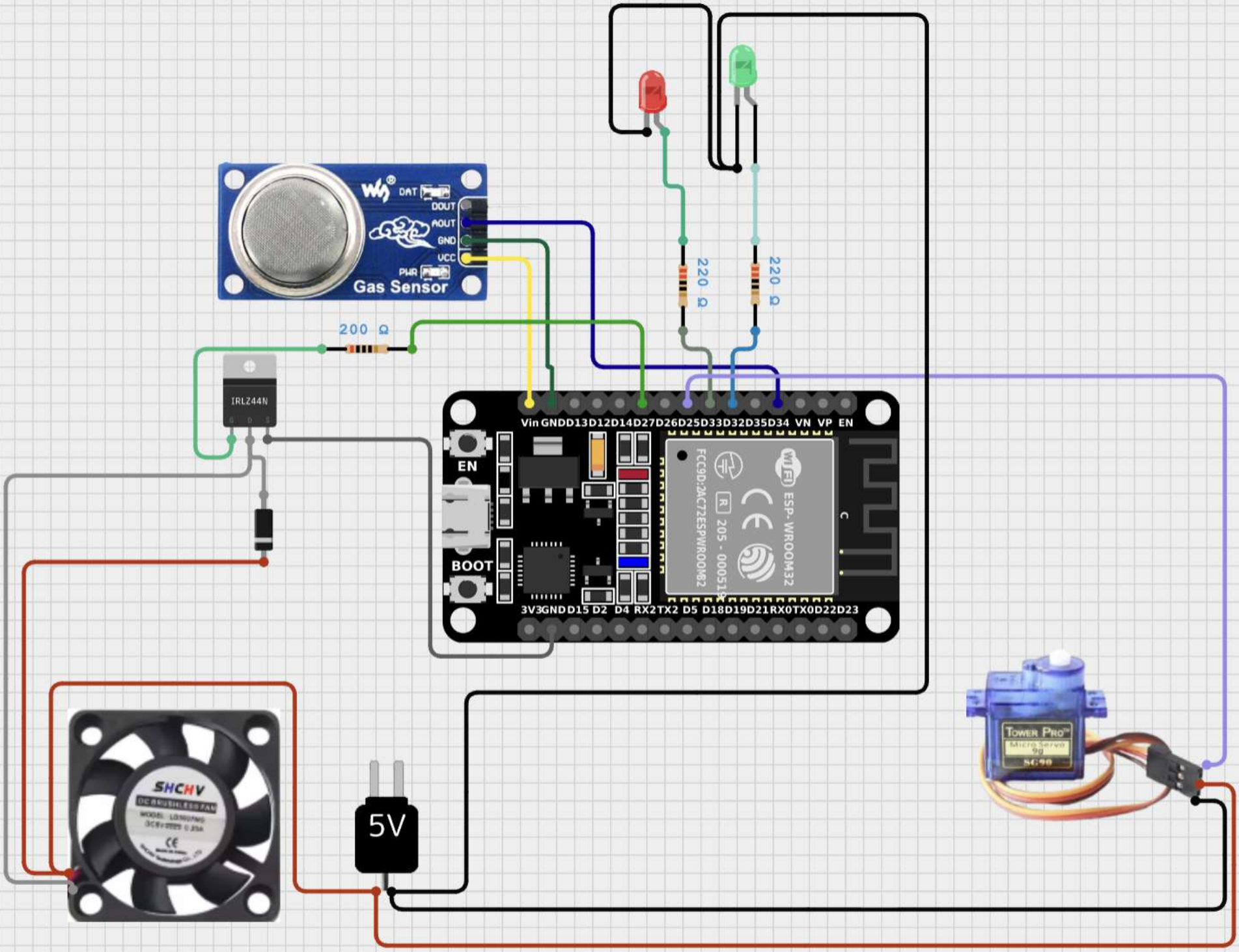
Acts as an electrically controlled switch, allowing the microcontroller to safely operate high-power devices like the exhaust fan.
Uses an electromagnetic coil to mechanically switch high-current circuits using a low-voltage control signal from the microcontroller.

Circuit Diagram



ESP32 Pin	Direction	Connected To
GPIO 34	Input (ADC)	MQ-6 Analog Ou (AO)
GPIO 25	Output (PWM)	Servo Motor Signal
GPIO 26	Output (Digital)	Active Buzzer
GPIO 27	Output (Digital)	MOSFET Gate (Fan)
3.3V	Power	MQ-6 VCC, Serv VCC
GND	Ground	Common GND for a modules
Vin/USB	Power Input	External 5V / USB

ESP32 Pin Allocation


DAT
DOUT
AOUT
GND
VCC
PWR
Gas Sensor
220 Ω
220 Ω
200 Ω
IRLZ44N
Vin GND D13 D12 D14 D27 D26 D25 D33 D32 D35 D34 VN VP EN
EN
BOOT
3V3 GND D15 D2 D4 RX2 TX2 D5 D18 D19 D21 RX0 TX0 D22 D23
ESP-WROOM32
5V
SHCHV
Tower Pro
Micro Servo
9g
SG90

ARDUINO IDE + ESP32

SOFTWARE IMPLEMENTATION

Development Environment

Arduino IDE with ESP32 board package. Required libraries:

`WiFi.h`	— Wi-Fi connectivity
`HTTPClient.h`	— REST API calls
`ESP32Servo.h`	— Servo control
`BlynkSimpleEsp32.h`	— Blynk integration

Core Logic:

Threshold-based detection using analogRead(). If gas_value > THRESHOLD, all safety routines execute sequentially before HTTP requests are made.

```
// Threshold Detection Core

int gasVal = analogRead(GAS_PIN);
if (gasVal > THRESHOLD) {
  digitalWrite(BUZZER, HIGH);
  digitalWrite(FAN, HIGH);
  servo.write(90); // Close valve
  // Upload to ThingSpeak
  sendToThingSpeak(gasVal);
  // Trigger Blynk event
  Blynk.logEvent("gas_alert");
  // Twilio SMS
  sendTwilioSMS();
}
```

CLOUD CONNECTIVITY

IOT PLATFORM INTEGRATIONS

~ ThingSpeak

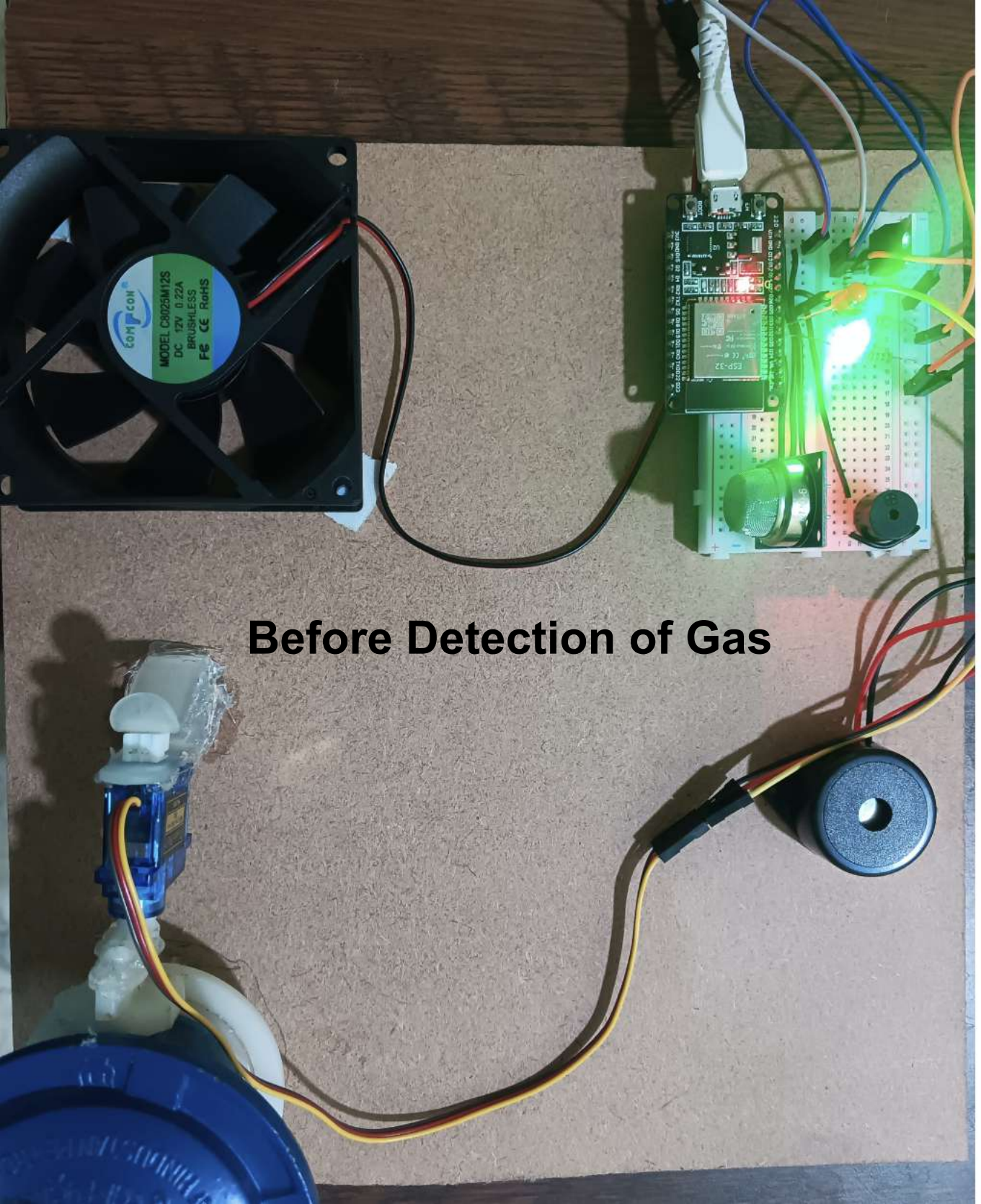
- ESP32 sends HTTP GET to ThingSpeak API with gas sensor value
- Data logged in Field 1 every 15 seconds continuously
- Built-in MATLAB visualisation generates real-time line graphs
- Enables trend analysis and historical data review

B Blynk IoT

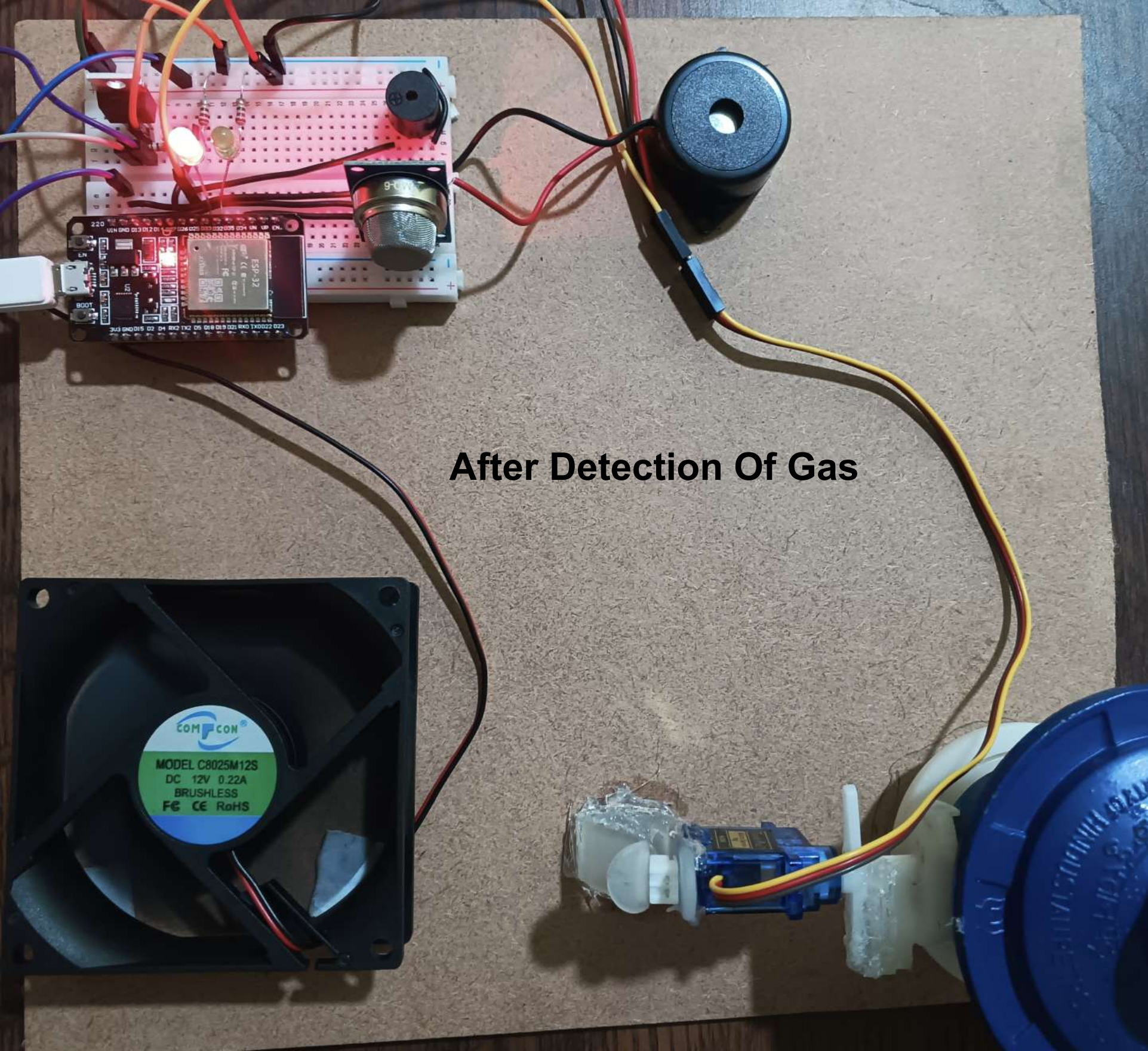
- ESP32 connects to Blynk Cloud via Auth Token
- Event "gas_alert" triggered on leakage detection
- Push notification delivered instantly to owner's mobile app
- Virtual pins display live sensor readings on dashboard

T Twilio SMS

- ESP32 makes HTTPS POST to Twilio REST API endpoint
- Message: "LPG Gas Leak Detected! Valve closed. Evacuate!"
- Delivered to pre-registered emergency contact number
- Works even when Blynk app is closed or offline



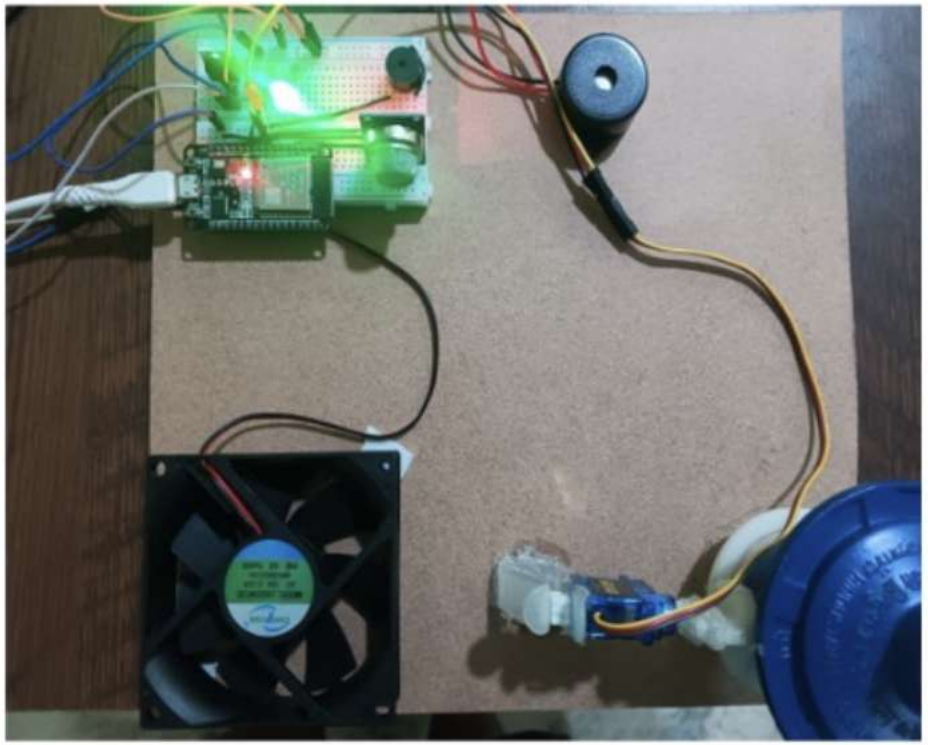
Before Detection of Gas


After Detection Of Gas
MODEL C8025M12S
DC 12V 0.22A
BRUSHLESS
RoHS

No spurious actuations occurred. MQTT reporting was uninterrupted throughout, with status "SAFE" confirmed on the mobile dashboard.

The clean-air baseline of 285 ± 42 ADC counts on the MQ-6 (compared to a typical MQ-2 baseline of 380 ± 90 ADC counts in the same environment) confirms the MQ-6's more stable response in ambient air — a consequence of its narrower gas-sensitivity profile reducing the response to background volatile organic compounds.

Interface in Thingspeak before Detection of gas



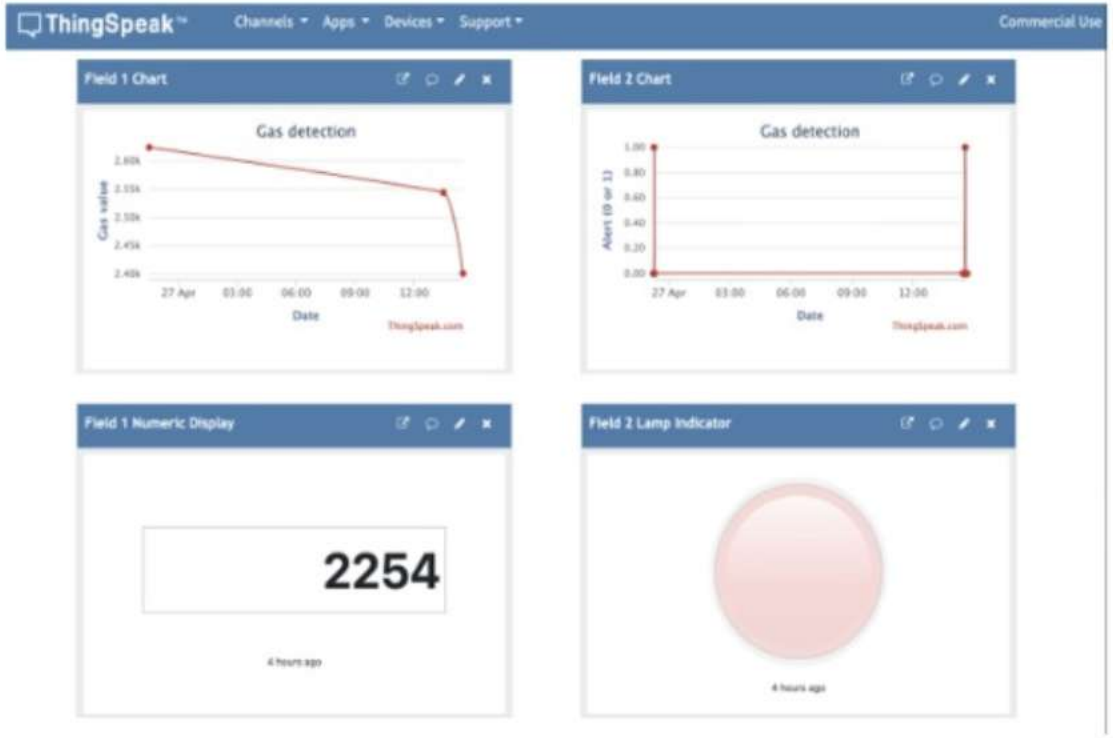
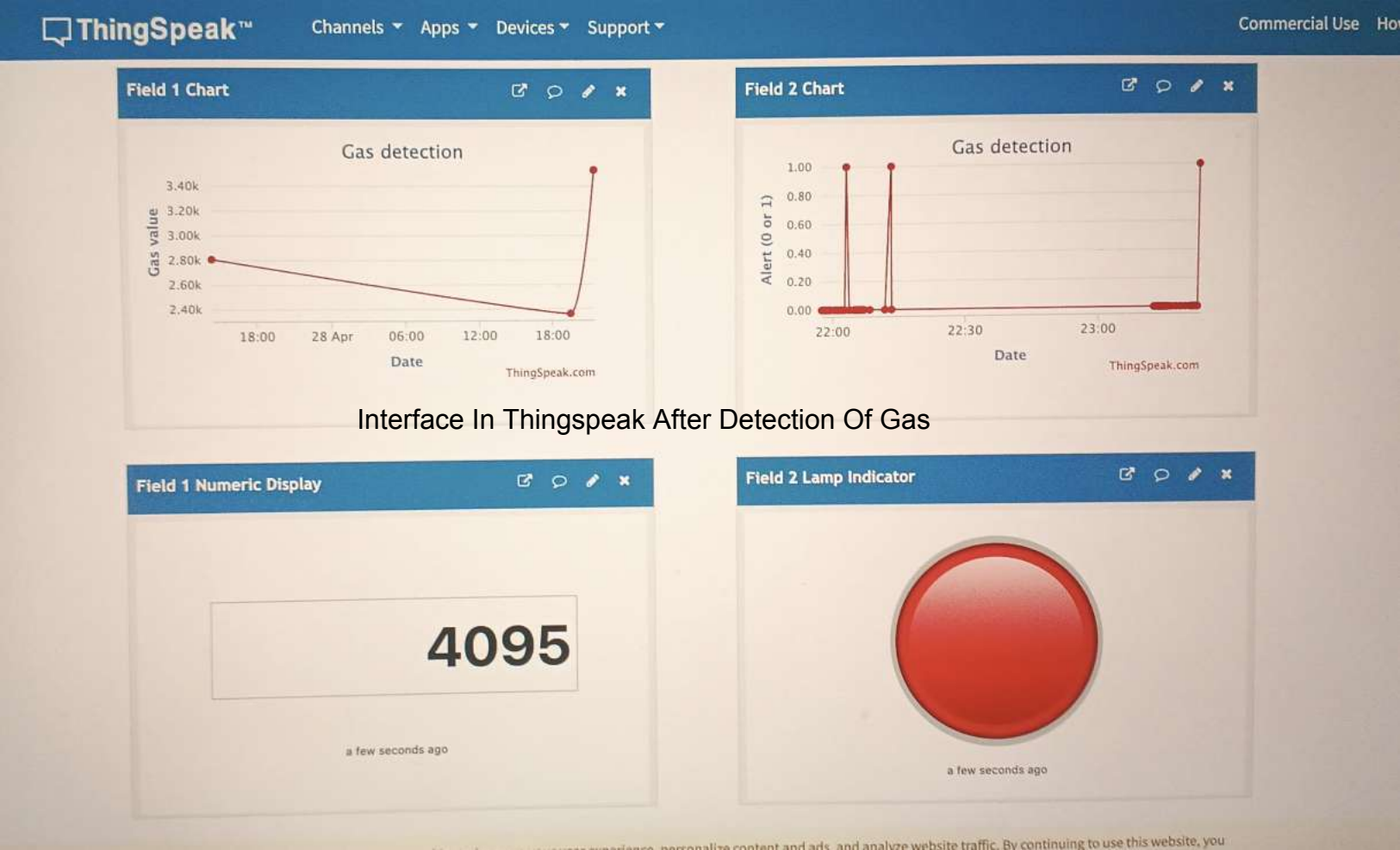
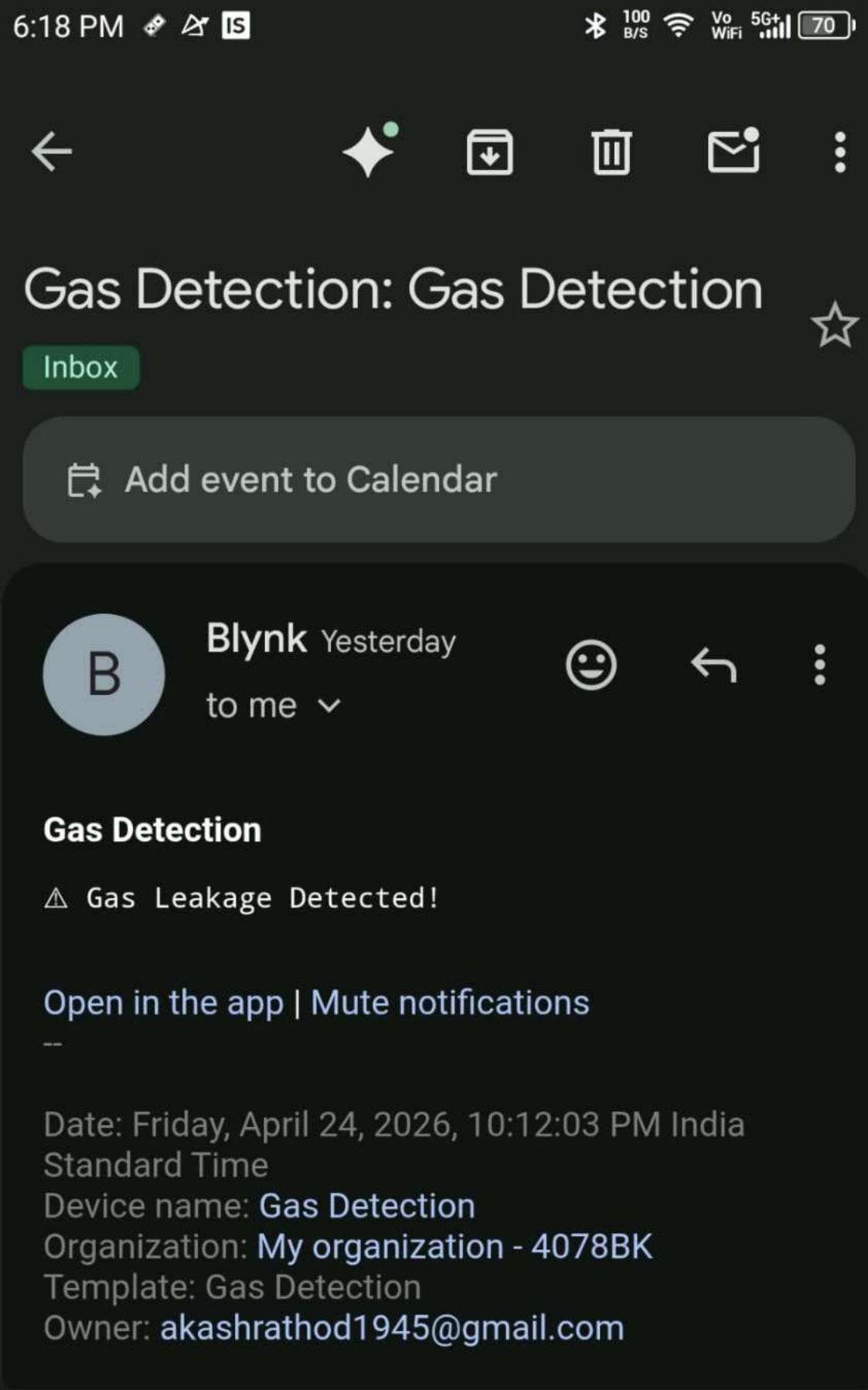
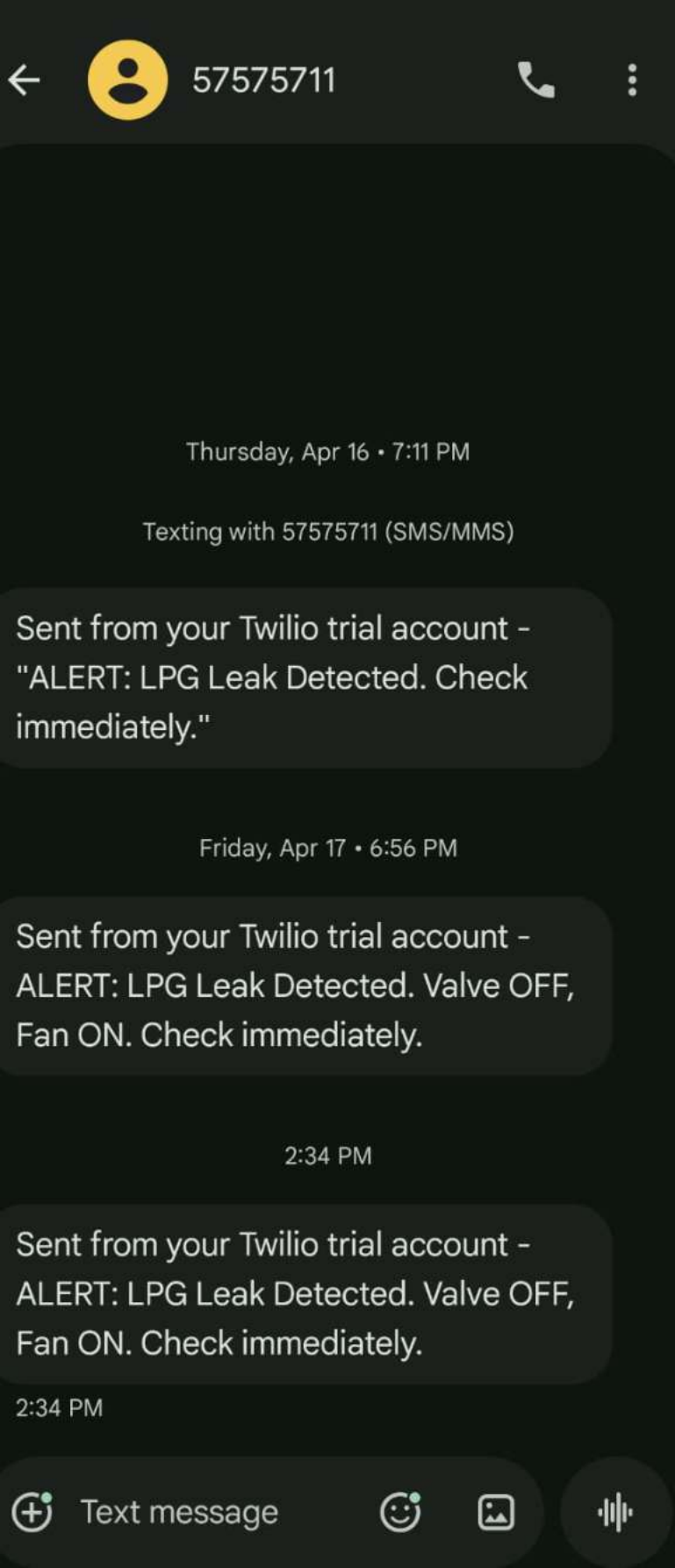


Figure 6.1: Serial Monitor Output — SAFE State (ADC readings, 500 ms interval)



Interface In Thingspeak After Detection Of Gas


6:18 PM
Gas Detection: Gas Detection
Inbox
Add event to Calendar
B
Blynk Yesterday
to me
Gas Detection
⚠ Gas Leakage Detected!
Open in the app | Mute notifications
--
Date: Friday, April 24, 2026, 10:12:03 PM India Standard Time
Device name: Gas Detection
Organization: My organization - 4078BK
Template: Gas Detection
Owner: akashrathod1945@gmail.com


57575711
Thursday, Apr 16 • 7:11 PM
Texting with 57575711 (SMS/MMS)
Sent from your Twilio trial account - "ALERT: LPG Leak Detected. Check immediately."
Friday, Apr 17 • 6:56 PM
Sent from your Twilio trial account - ALERT: LPG Leak Detected. Valve OFF, Fan ON. Check immediately.
2:34 PM
Sent from your Twilio trial account - ALERT: LPG Leak Detected. Valve OFF, Fan ON. Check immediately.
2:34 PM
Text message

Gas Detection •



Exhaust Fan

OFF

Servo Moto

OFF